

Andrew Murwin

andrew.murwin@gmail.com | 480-309-1214 | andrewmurwin.com

Professional Experience

Software Engineer II, Starbucks – Phoenix, AZ May 2023 – Present

- **Microfrontend architecture:** Implemented React MFE superapp using Webpack Module Federation to decouple three existing web apps, improving feature release velocity by 20% through eliminating testing and deployment bottlenecks
- **Unified Platform:** Implemented and presented React Native Greenfield mobile app unification platform to cross-functional stakeholders and leadership. Provided tradeoffs for governance, security, adoptability and component ownership to stakeholders and secured alignment on tech stack and approach
- **Webpack v4 → v5 Migration:** Led codebase migration of 10 modules to Webpack 5, reducing dev server deployment build time by 40% and improving stability by 30%, and enabling adoption of Microfrontend architecture for three teams post-migration
- **Design System:** Led design token alignment effort on 500+ components to facilitate faster design to functional frontend conversion. First modal redesign implementation led to a 80% reduction in new modal implementation time
- **Routing Standardization:** Designed and drove org-wide adoption of a React Router search parameter pattern that auto-standardizes parameter validation and default values leading to reduced bugs and improving routing feature implementation time by 70%
- **Testing Infrastructure:** Designed and shipped a standardized Jest test wrapper that fixed the most common developer testing gaps and established shared testing patterns at scale, reducing test development time by 30%

Web Development Intern, Starbucks – Phoenix, AZ May 2022 – Aug 2022

- Designed and implemented a Role-Based Access Control (RBAC) feature using React and GraphQL; enhanced application accessibility with ARIA labels and improved color contrast, leading to 90% WCAG 2.1 AA compliance

Software Engineer Intern, Comtech Telecommunications – Tempe, AZ Dec 2021 – May 2022

- Reduced time spent locating project contribution data by 60% by implementing a full-stack MVC dashboard with C# LDAP integrations against Active Directory

Business Data Analyst (Freelance), Mindful Planet – Phoenix, AZ Jan 2018 – Jan 2025

- Automated Excel reporting using Python (pymongo, pandas, xlsxwriter) to extract and transform data from a MongoDB database, establishing a reusable data pipeline for report generation and documenting the shared database structure for contractor onboarding

Projects

Crypto Trading Platform - [GitHub](#)

- Full-stack algorithmic trading system with a FastAPI/GraphQL backend, exchange adapters, and a server side rendered React dashboard for real-time portfolio monitoring and P&L analysis

The Efficacy of Different Timesteps in Data when Predicting Cryptocurrency Prices - [GitHub](#)

- Investigated ML model performance for crypto price forecasting using a Python/pandas/PyTorch pipeline

Skills

Backend & Data: Node.js, PostgreSQL, MongoDB, Python, pandas, PyTorch, .NET/C#

Frontend: React, React Router, GraphQL, Apollo Client, TypeScript, JavaScript, HTML5, CSS3, Webpack

Testing & Tooling: Jest, React Testing Library

Education

MS, Computer Science (Big Data) | Arizona State University | 4.0 GPA Dec 2023

- Graduate Coursework: Database Management System Implementation, Distributed Database Systems, Statistical Machine Learning, Data Mining, Semantic Web Mining, Knowledge Representation and Reasoning

BS, Computer Science | Arizona State University | Barrett, The Honors College | 4.0 GPA Dec 2022